Scrollytelling and statistical inference

Overview

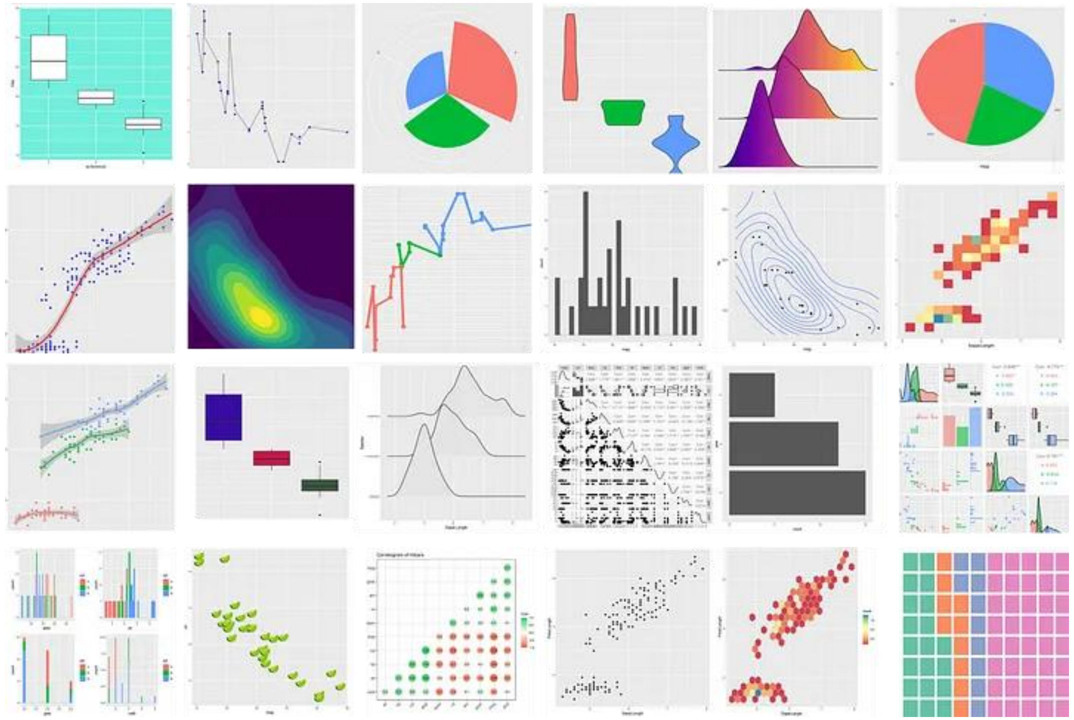
A few last topics in ggplot

- Quick review of the grammar of graphics
- ggplot bonus features

Scrollytelling with Closeread

If there is time

- A brief overview of statistical inference
- Hypothesis tests



Homework 4

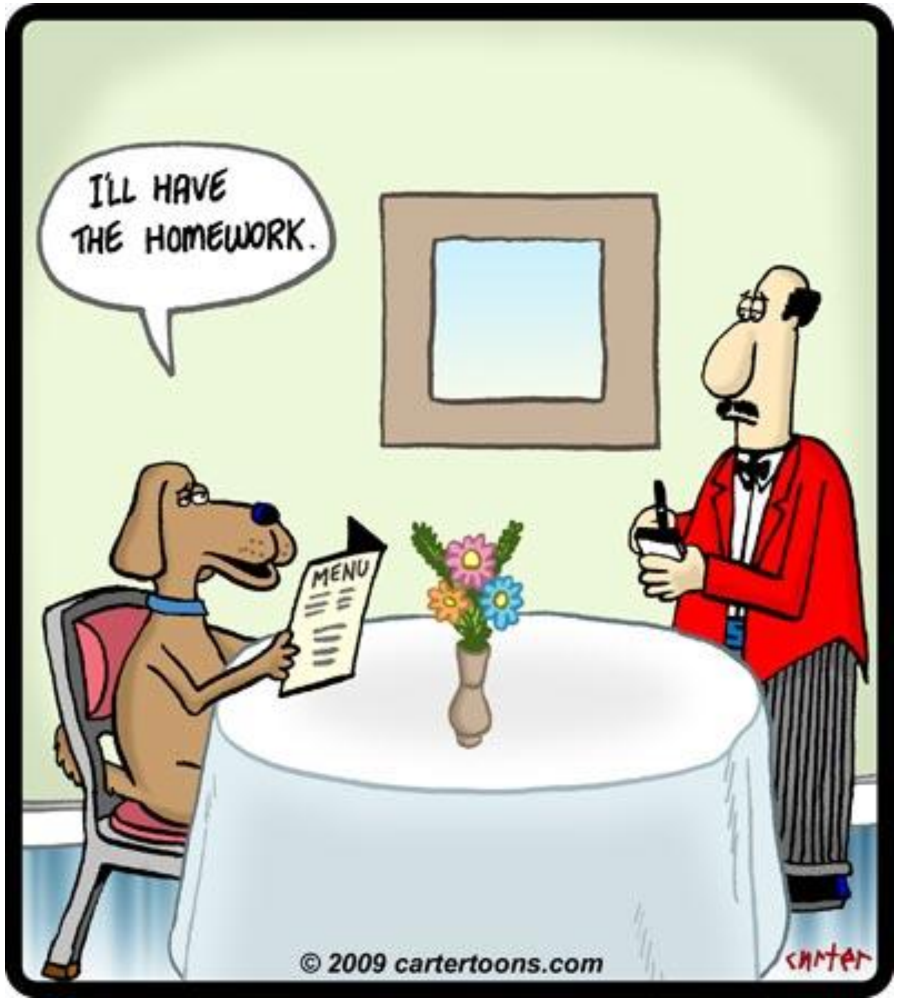
Create a scrollytelling webpage where you analyze a data set of your choosing

Homework 4 plan:

- Draft of homework due by 11pm on Monday July 27th
- Final version is due by 10pm on Wednesday July 29th

You will upload a quarto document and a hmtl document to Canvas

How did homework 3 go?



Class exam

Tuesday July 28th in person during regular class time

- Exam is on paper

If you have accommodations for extra time, stay in the classroom and keep working on the exam

Exam “cheat sheet”

You are allowed an exam “cheat sheet”

One page, single-side, that contains **only code and equations**

- No code comments allowed
- E.g., plot(x, y)

Cheat sheet must be on a regular 8.5 x 11 piece of paper

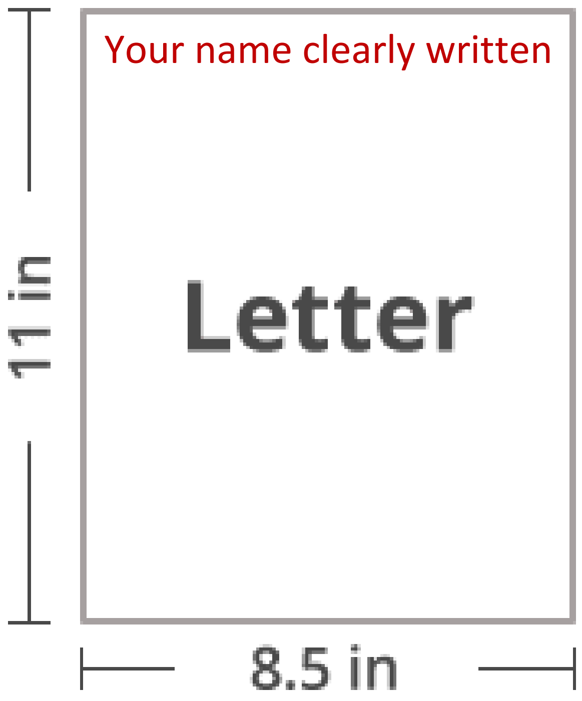
- Your name on the upper left side of the paper

I recommend making a typed list of all functions discussed in class and on the homework

- This will be useful beyond the exam

You must turn in your cheat sheet with the exam

- Failure to do so will result in a 20 point deduction



Questions?

Review and continuation of ggplot

Review: Graphs are composed of…

A Frame: Coordinate system on which data is placed

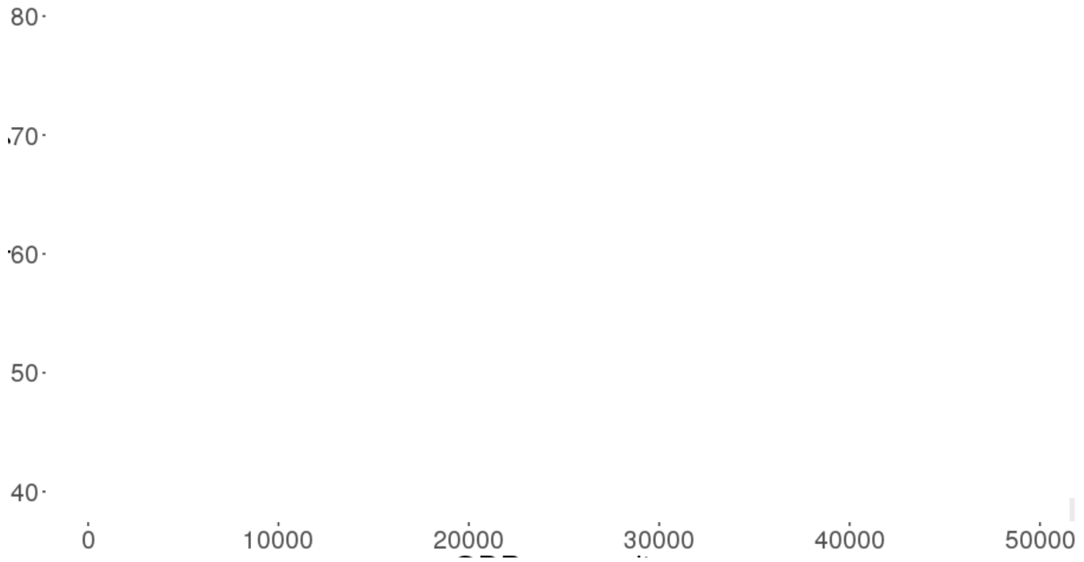
- ggplot() +

Glyphs: basic graphic unit representing cases or statistics

- Data is **mapped** onto these aesthetics such as: shape, color, size, etc. and/or aesthetics can be set to a fixed value
    - geom_point(aes(x = gdpPercap, y = lifeExp, color = continent)) geom_point(aes(x = gdpPercap, y = lifeExp), color = "red")

Scales and guides: shows how to interpret axes and other properties of the glyphs

- scale_x_continuous(trans = "log10") scale_color_brewer(type = "qua", palette = 2)



Review: Plots can also contain…

Facets: allows for multiple side-by-side graphs based on a categorical variable

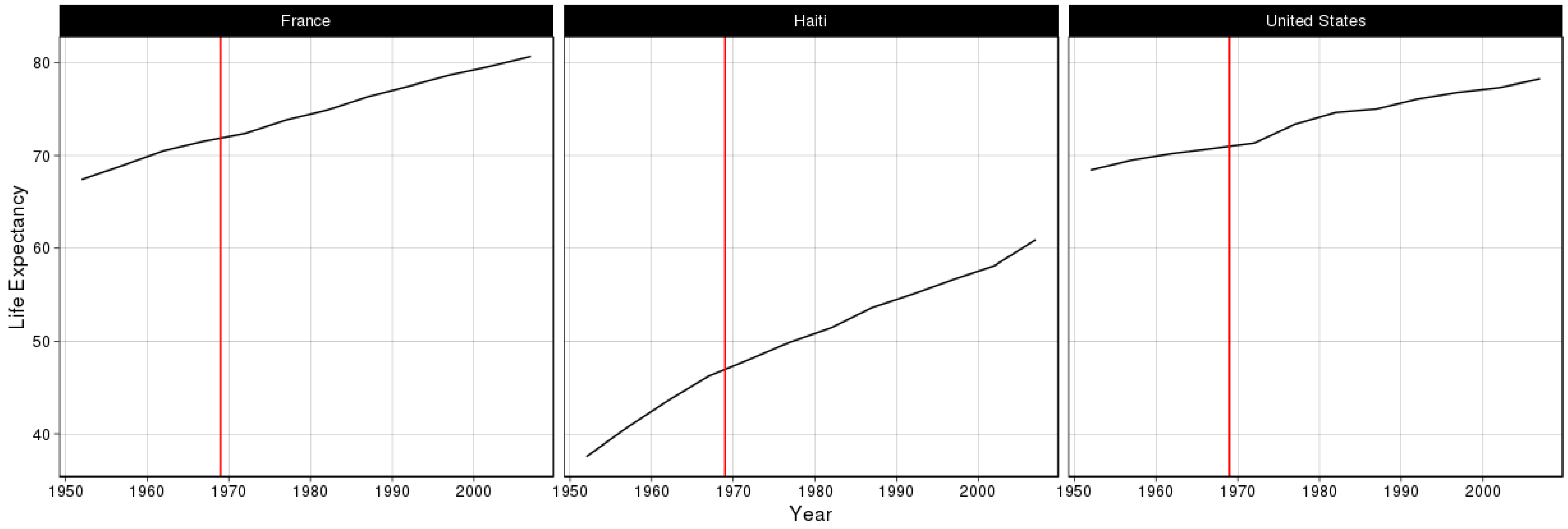
- facet_wrap(~country)

Layers: allows for more than one types of data to be mapped onto the same figure

- geom_vline(xintercept = 1969, col = "red")

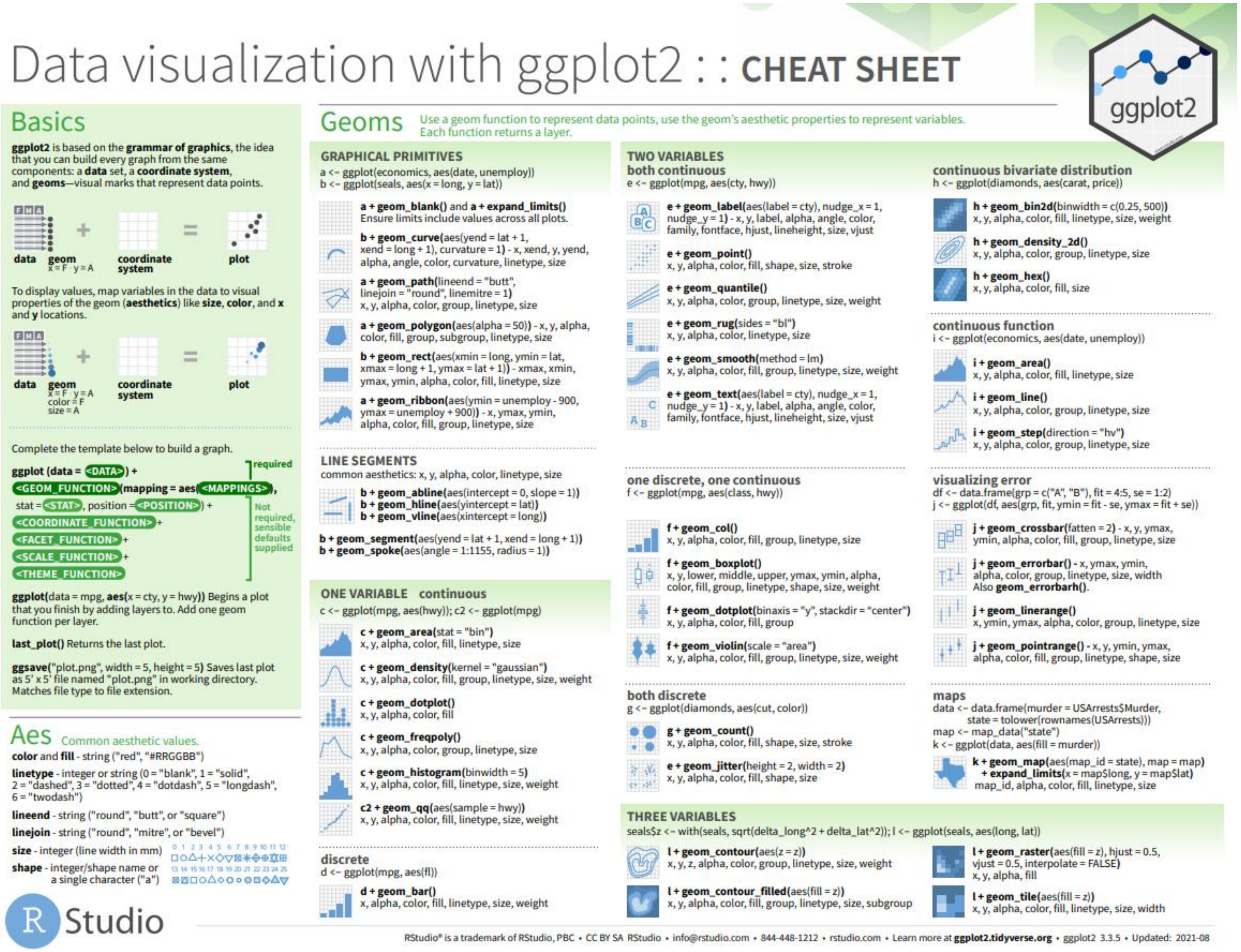
Theme: contains finer points of display (e.g., font size, background color, etc.)

- theme_wsj()



Questions?

Do you want to go over a few last features of ggplot by continuing with the class 6 code?



Scrollytelling with Closeread

Scrollytelling

"Scrollytelling" is a web design technique that uses visual and textual elements to tell a story as the reader scrolls through a page

- Closeread examples, and other examples

We can create scrollytelling documents by using the Closeread custom format in Quarto

To use Closeread, you need to install the Quarto extension by running the following at the terminal tab:

quarto add qmd-lab/closeread

This create a directory called _extensions which has the code necessary to make a Closeread webpage

Creating a Closeread Quarto document

To create a Closeread Quarto document, we start by specifying the appropriate quarto header

```
---
title: My Closeread story
format: closeread-html
---
```

```
---
title: "Class 25 notes and code"
format: pdf
---


<!--  Please run the code in the  R chunk below once. This will install some packages and download data and images needed for these exercises.   -->


```{r message=FALSE, warning = FALSE, echo = FALSE, eval = FALSE}

SDS230::download_data("IPED_salaries_2016.rda")

```



```{r setup, include=FALSE}

# install.packages("latex2exp")

library(latex2exp)
library(dplyr)
library(ggplot2)

#options(scipen=999)


knitr::opts_chunk$set(echo = TRUE)

set.seed(123)

```

```

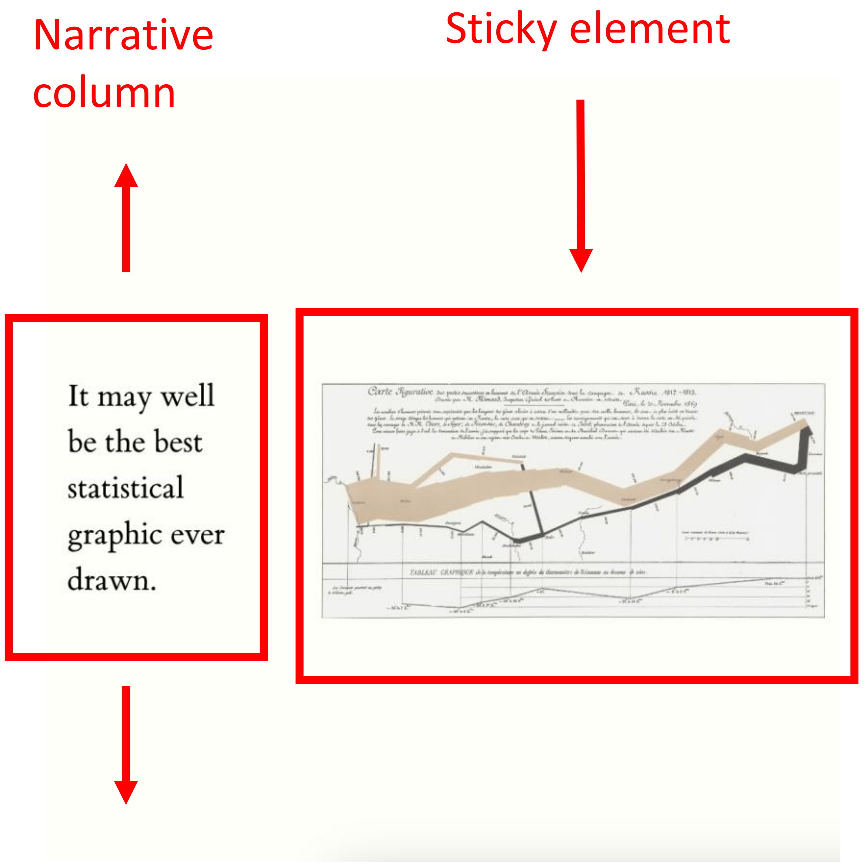
# Closeread components

Every Closeread document consists of **sections** of the document flagged as Closeread sections

Within each Closeread section:

1. “**sticky**” **elements** are tagged to appear in the main column of the Closeread section

2. “**narration”** text appears in the “narrative column” and triggers the appearance of the sticky element



# Closeread sections

Closeread sections specify what is part of document is a scrollingtelling story

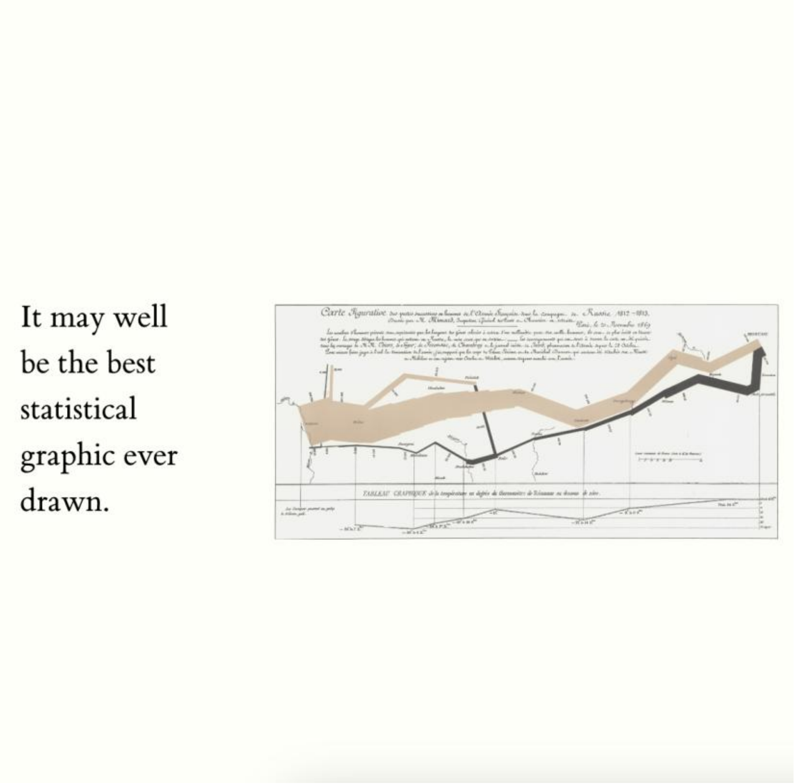
- Elements outside of a section will appear as a normal text
    - (i.e., as a normal webpage/HTML document)

::::{.cr-section}

Text and images/code go in here…

::::

Closeread sections must have at least 3 colons.
They can have more colons to make sections stand out.



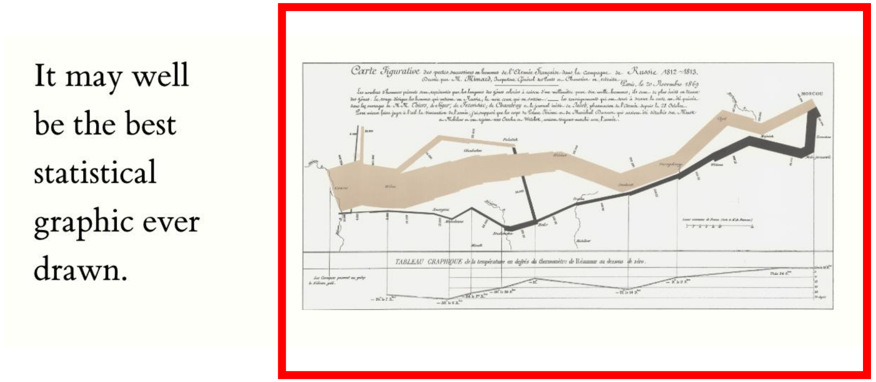
# Adding stickies

Elements that are pinned as one scrolls are called ”stickies”

We can add stickies using:

```
:::{#cr-stickyname}
        sticky content
:::
```

Must start with #cr-




```
:::{#cr-myimage}
    ![](path-to-image.png)
:::
```

```
:::{#cr-myplot}
    ```{r}
        hist(rnorm(15))
    ```
:::
```

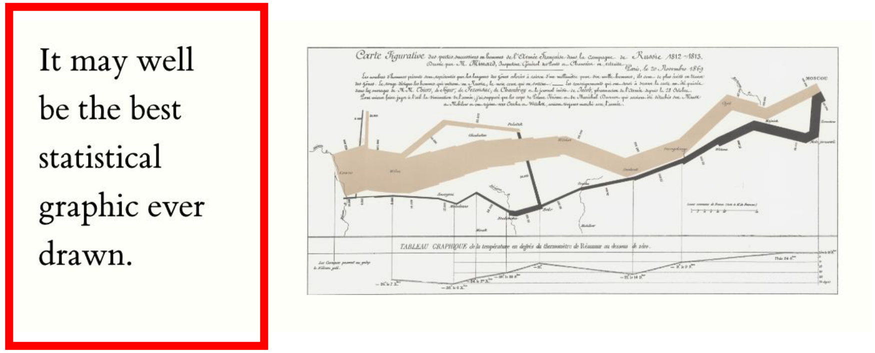
Triggers

Any text (or other elements) in a Closeread section that are not stickies will be placed in the narrative column

You can set a "narrative text" to trigger focus on a particular sticky by using: @cr-mysticky

It may well be the best statistical grapic every drawn

@cr-myimage



Let's now show a histogram
@cr-myplot

Focus effects

We can add “focus effects” which can do the following:

- Scale, pan, or zoom in on images (or other elements)
- Highlight lines of text

Example:

```
This is where we load the library. [@cr-dplyr]{highlight="1"}
This calculates summary statistics. [@cr-dplyr]{highlight="2-3"}

:::{#cr-dplyr}
    ```{r}
    library(dplyr)
    group_by(mtcars, am) |>
        summarize(avg_wt = mean(wt))
    ```
:::
```

Layouts

We can also change the layout of where the narration text and stickies appear by modifying the Quarto header

```
---
  format:
    closeread-html:
      cr-section:
        layout: "overlay-center"
---
```

Options are:

- sidebar-left   sidebar-right
- overlay-right   overlay-right   overlay-center

Let’s try it in R!

SDS1110:download_class_code(7)

Also see [this app](#) which can help create Closeread documents

A brief introduction to Statistical inference

Statistical Inference

In **statistical inference** we use a sample of data to make claims about a larger population (or process)

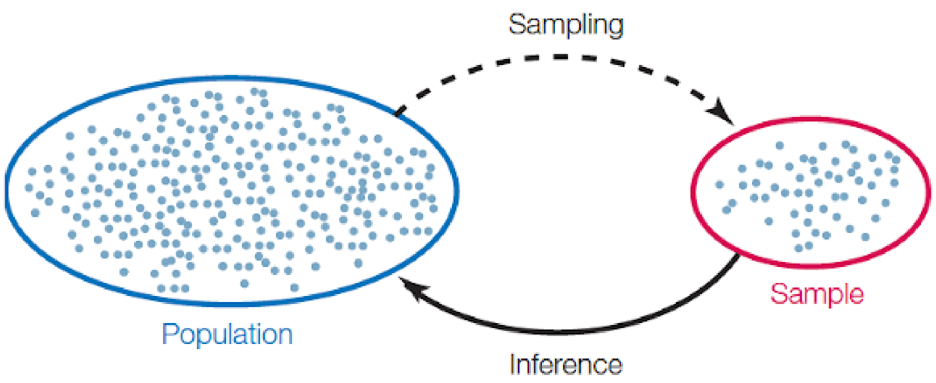
Examples:

<u>OkCupid data</u>

- **Sample**: 59,000 OkCuipd users from San Francisco
- **Population**: What is the population of interest?

<u>Billionaire data</u>:

- **Sample**: 2,781 billionaires in 2024
- **Population**: What is the population of interest?



Population: all individuals/objects of interest

Sample: A subset of the population

Terminology

A parameter is number associated with the <u>population/process</u>

- e.g., population mean height of all men μ

A **statistic** is number calculated from the <u>sample</u>

- e.g., the height of a sample of n = 59,000 men $\bar{x}$

A statistic can be used as an estimate of a parameter

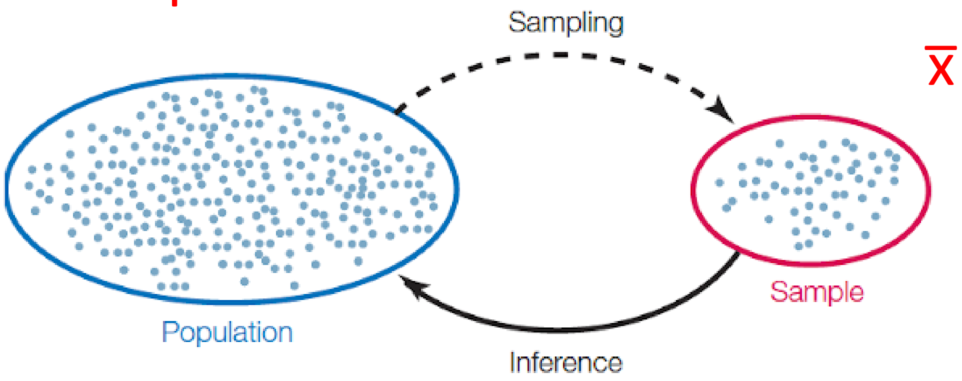
- A parameter is a single fixed value
- Statistics tend to vary from sample to sample

parameter

μ

statistic

$\bar{x}$



Example:

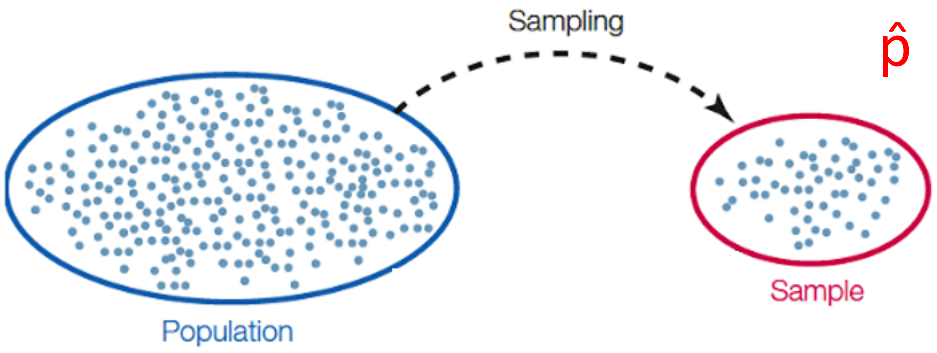
- Using the mean height of 59,000 men ($\bar{x}$) to estimate the mean height of all men (μ)

Examples of parameters and statistics

	Sample statistic	Population parameter	Example from class
Mean	$\bar{x}$	μ	mean(profiles$height) $\bar{x} = 68.3$
Proportion	$\hat{p}$	π	prop.table(table(profiles$drinks))[6] $\hat{p} = 0.697$
Correlation	r	ρ	cor(min_temp, max_temp) $r = 0.967$

Sampling

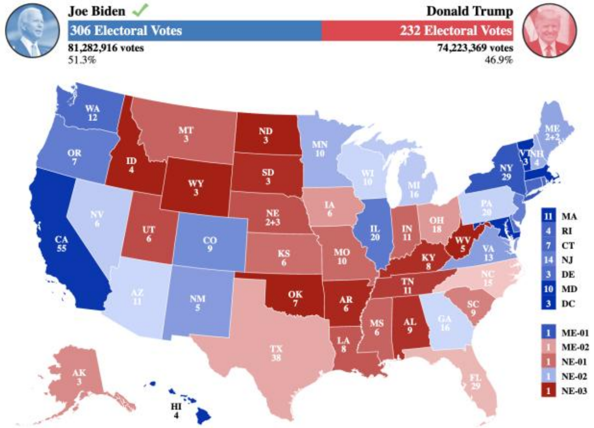
parameter π



statistic $\hat{p}$

Simple random sample: each member in the population is equally likely to be in the sample

- Allows for generalizations to the population



Polls of 1,000 voters: $\hat{p}_{\text{Trump}}$

Vote on election day: π_{Trump}

Hypothesis tests

Statistical tests   (hypothesis test)

A **statistical test** uses data from a sample to assess a claim about a population (parameter)

<u>Example 1</u>: The average body temperature of humans is 98.6˚

How can we write this using symbols?

- $\mu = 98.6$

Statistical tests (hypothesis test)

A **statistical test** uses data from a sample to assess a claim about a population (parameter)

<u>Example 2</u>: More than half of voters disapprove of Trump's job performance

How can we write this using symbols?

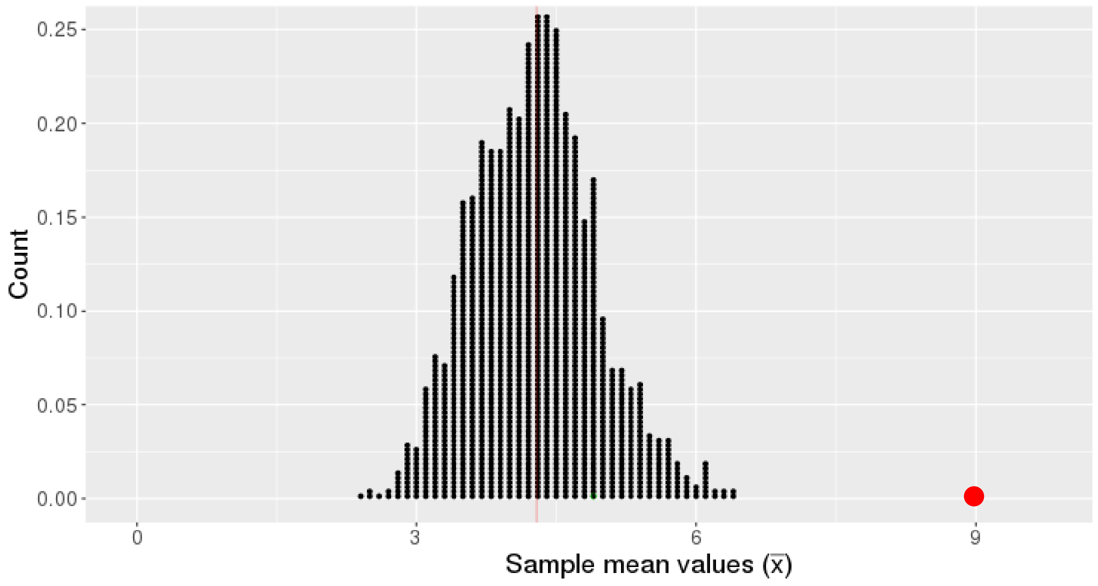
- $\pi_{\text{Disapprove}} > 0.5$

Basic hypothesis test logic

We start with a claim about a population parameter

- E.g., $\mu = 4$

This claim implies we should get a certain distribution of statistics



If our observed statistic is highly unlikely, we reject the claim

Example claims (hypotheses)

Let’s see if we can write the following claims (hypotheses) using symbols

Claim: 88% of Yale students graduate within four years

Claim: The average age of a Yale undergraduate is 20

Claim: 70.7% of Yale classrooms have fewer than 20 students in attendance

Testing claims (hypotheses)

Claim: 88% of Yale students graduate within four years

- H: $\pi = 0.88$

- To test this claim, we could randomly selected n = 100 Yale graduates

- If we found the proportion that graduated in 4 years is $\hat{p} = .80$, would we believe the claim?

Testing claims (hypotheses)

Claim: The average age of a Yale undergraduate is 20

- H: $\mu = 20$
- To test this claim, we could randomly selected n = 50 Yale graduates
- If we found the average age of in our sample of students was $\bar{x} = 20.2$, would we believe the claim?

Visual hypothesis test

Visual hypothesis test

In visual hypothesis tests, we create data visualizations to try to assess whether particular relationships exist in our data

- One way this is done through a visual lineup

Let’s try it in R!